

Week 5 · Waypoint Following and LOS Guidance

USV Guidance, Navigation and Control (Graduate) · Department of Autonomous Vehicle System Engineering, Chungnam National University | revised 2026-09-22

★ Reference material — read this first

The five courses below are **taught by the instructor of this course** and are the assumed background for it. They run from the fundamentals down to the graduate material, so start wherever the gap is. Every frame, symbol and derivation used here is developed in them at length; anyone whose prerequisites are thin should work through them first, then return.

#	Course	Level	Lang.	Video	Slides and code
1	Control Engineering (제어공학) — the foundation: transfer functions, feedback, stability, root locus, PID	undergraduate	KO	playlist	drive
2	Control System Design (제어시스템설계) — design rather than analysis: specifications, loop shaping, discrete implementation	undergraduate	KO	playlist	drive
3	Advanced Control Engineering (제어공학특론) — reference frames, the 6-DOF equation of motion, rotation matrices and Euler angles, linearisation and trim, vehicle control design	graduate	EN	playlist	drive
4	Sensor Signal Processing and Fusion (센서신호처리 및 융합) — sensor models, noise, estimation and multi-sensor fusion	graduate	EN	playlist	drive
5	Capstone Design (캡스톤디자인) — a vehicle project carried end to end	undergraduate	KO	playlist	drive

Not fluent in MATLAB or Simulink yet? Do these before Part 2. Every laboratory in this course is Simulink, and the Onramp courses are free and take a few hours each.

Tool	Where to start
MATLAB	MATLAB Onramp · Core MATLAB Skills
Simulink	Simulink Onramp · instructor's Simulink lectures part 1 · part 2

🔧 Getting the course files, and keeping them current (Windows)

The notes, models and scripts are kept in one Git repository that is updated through the semester. Clone it once; before every class, pull. A pull downloads only what has changed since the last one.

When	Where to run it	Command
once	PowerShell — installs Git for Windows	<code>winget install --id Git.Git -e</code>
once	the folder that will hold the course, e.g. <code>Documents</code>	<code>git clone https://github.com/wkyouncnu/Sensor-Signal-Processing-and-Fusion.git</code>
before every class	inside the cloned folder <code>Sensor-Signal-Processing-and-Fusion</code>	<code>git pull</code>

- `git pull` prints `Already up to date.` when nothing has changed, and otherwise lists the files it updated.
- The repository is private. When Git asks, sign in with the GitHub account the instructor has given access.
- Experiment on copies, not on the cloned files: copy a week's `WXX_simulink` folder elsewhere first, and a pull can never collide with local edits. If it already has, `git stash`, then `git pull`, then `git stash pop` sets the edits aside, updates, and puts them back.
- The MSS toolbox is not part of the repository. The weeks that simulate the Otter need it at `Tools\MSS` inside the cloned folder.

- **Course:** USV Guidance, Navigation and Control (Graduate)
- **Department:** Autonomous Vehicle System Engineering, Chungnam National University
- **This week:** ① where the heading command comes from: guidance turns a list of waypoints into ψ_d ② line of sight (LOS) as a P controller on the cross-track error, tuned by its look-ahead distance ③ when to move to the next leg ④ a current, and the integral that removes the offset it leaves (ILOS) — all tuned from measurements, with no model of the vessel

★ Prerequisites from the previous week

- Week 4 tuned a heading autopilot on the Otter. This week puts a guidance law in front of it and does not retune it: the autopilot follows whatever heading it is given.
- Week 4 §4-4: heading is not course. A vessel in a current points one way and travels another; §5-5 is where that difference costs a path error.
- From Weeks 2 to 4: P speeds a loop up and leaves an error when something pushes back; I removes that error; the tuning order.

Learning Outcomes

Upon completion of this week, the learner is able to:

1. Compute the cross-track error of a vessel from the two waypoints of a leg, and explain why aiming at a waypoint does not follow the line between waypoints.
2. Write the LOS law, recognise it as a P controller on the cross-track error with gain $1/\Delta$, and choose Δ from a measured sweep.
3. Choose the switching distance from the corner behaviour it produces on a mission.
4. Predict the offset a cross current leaves under LOS, $\Delta \tan$ of the heading held, and remove it with ILOS.
5. Tune a guidance law by the order $\Delta \rightarrow R \rightarrow \kappa$, stating the measurement behind each choice.

Prerequisites and Setup

Item	Requirement
MATLAB	R2024b, with Simulink
MSS	<code>Tools/MSS</code> , found by <code>W05_0_setup</code> through <code>mss_path</code>
Course folder	<code>lectures/W05_simulink</code>
Models	one per experiment — <code>W05_C_atan2</code> , <code>W05_D_LOS</code> , <code>W05_E_switching</code> , <code>W05_F_LOS</code> , <code>W05_F_ILOS</code> , <code>W05_G_tuning</code> — all generated by <code>W05_1_build_guidance</code> , never edited by hand
Time	lecture 40 min, laboratory 1 h 20 min

Part 1 · Principle and experiment, section by section

Every section states what is observed, says what follows from it, and then runs **the experiment that measures it**, on its own model. The numbers quoted in the text are printed by the run a few lines below it.

Part of a section	What it holds
What is observed	the effect itself, with the numbers it produces
What follows from it	numbered lines, one step each, from the geometry of the path to the law
Experiment N-x	the model, the commands, the output actually obtained, the figure, and a table that reads every feature of the figure back to the numbered line that predicts it

5-0. Setting up (10 min)

```
cd lectures/W05_simulink
W05_0_setup
W05_1_build_guidance
```

Expected output:

```
W05_0_setup
  path      5 waypoints, legs of 60 m
  guidance  Delta = 5 m (P gain 0.200), R_switch = 3 m, kappa = 0.3
  start     20 m east of the path; current 0 m/s

built W05_C_atan2.slx      (overlapping lines: 0)
built W05_D_LOS.slx       (overlapping lines: 0)
built W05_E_switching.slx (overlapping lines: 0)
built W05_F_LOS.slx       (overlapping lines: 0)
built W05_F_ILOS.slx      (overlapping lines: 0)
built W05_G_tuning.slx    (overlapping lines: 0)
```

- `W05_0_setup.m` is the only file edited by hand. If it stops in `mss_path`, the MSS toolbox is not at `Tools\MSS`.
- The first line of `W05_0_setup` is `clear`. Run it again whenever a Scope does not match these notes.

5-1. What guidance adds

This section answers: where does the heading command come from?

- Weeks 3 and 4 were given their commands by hand: a speed, a heading. A vessel on a mission is given **waypoints** instead, and something must turn them into a heading to steer. That something is **guidance**.
- Guidance is a controller around the heading autopilot. Its output, ψ_d , is the autopilot's input; its input is the vessel's position; its error is the distance from the path.

Every model this week has the same four blocks. Each is a MATLAB Function with commented code; double-clicking one shows the equations of this section.

Block	Input → output	This week
guidance	position → commanded heading ψ_d	the subject: atan2, LOS or ILOS
heading autopilot	ψ_d and the state → yaw moment N	the gains of Week 4, $K_p = 300$, $K_d = 100$, with P on the wrapped heading error and D on the measured yaw rate — so that the jump in ψ_d at a waypoint switch gives no derivative kick (Week 2 §2-10); not retuned
allocation	$N \rightarrow$ two shaft speeds	Week 4 §4-1, with a constant surge force $X_{ff} = 60$ N (about 0.77 m/s)
Otter	shaft speeds → twelve states	Week 1

The error guidance works on. "Follow the path" has to become a number before a controller can act on it, and **one number is not enough**. The path is a straight leg from waypoint k to waypoint $k+1$; what is needed is how far **along** it the vessel has come and how far **off** it the vessel is. Both come out of one construction.

Step 1 — the leg gives the frame its direction. The two waypoints fix a direction measured from north:

$$\pi_p = \text{atan2}(E_{k+1} - E_k, N_{k+1} - N_k)$$

- π_p is **constant along a leg** and is recomputed only when the waypoint index changes.
- It must be the two-argument **atan2**. A leg can run into any of the four quadrants, and the one-argument **arctan** folds two of them onto the other two — the same reason Week 4 §4-6 needs it.

Step 2 — the leg carries two unit vectors. Along the leg, and to its right:

$$\hat{\mathbf{t}} = \begin{bmatrix} \cos \pi_p \\ \sin \pi_p \end{bmatrix} \quad \hat{\mathbf{n}} = \begin{bmatrix} -\sin \pi_p \\ \cos \pi_p \end{bmatrix}$$

They are perpendicular and of unit length, so together they are a **frame** — the path frame $\{p\}$ — and $\hat{\mathbf{n}}$ points to starboard of a vessel running along the leg.

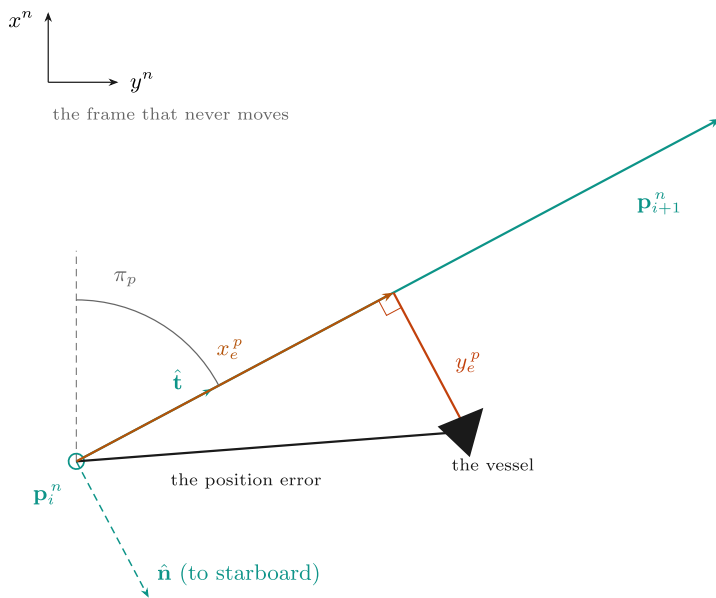
Step 3 — resolve the position error in that frame. Take the vector from the waypoint to the vessel and ask for its two components:

$$\begin{bmatrix} x_e \\ y_e \end{bmatrix} = \mathbf{R}(\pi_p)^T \begin{bmatrix} N - N_k \\ E - E_k \end{bmatrix}, \quad \mathbf{R}(\pi_p) = \begin{bmatrix} \cos \pi_p & -\sin \pi_p \\ \sin \pi_p & \cos \pi_p \end{bmatrix}$$

which written out is the pair this week uses everywhere:

$$\begin{aligned} x_e &= (N - N_k) \cos \pi_p + (E - E_k) \sin \pi_p \\ y_e &= -(N - N_k) \sin \pi_p + (E - E_k) \cos \pi_p \end{aligned} \quad \begin{array}{l} \text{along the leg} \\ \text{across it, positive to starboard} \end{array}$$

Symbol	Quantity	Unit · source
$(N_k, E_k), (N_{k+1}, E_{k+1})$	the two waypoints of the active leg	m · WP_N, WP_E
(N, E)	the vessel's position, north and east	m · states 7 and 8
π_p	the direction of the leg, from north	rad
x_e	along-track error: distance travelled along the leg	m · §5-4 switches on it
y_e	cross-track error: distance off the leg, positive to its right	m · §5-3 drives it to zero
$d = \ \mathbf{p}_{k+1} - \mathbf{p}_k\ $	the leg's length	m · $d - x_e$ is what is left to run



one rotation, two answers

The leg gives the frame its direction:

$$\pi_p = \text{atan2}(y_{i+1}^n - y_i^n, x_{i+1}^n - x_i^n)$$

and the error, written in that frame, is

$$\begin{bmatrix} x_e^p \\ y_e^p \end{bmatrix} = \mathbf{R}_p^n(\pi_p)^\top \begin{bmatrix} x^n - x_i^n \\ y^n - y_i^n \end{bmatrix}$$

x_e^p — how far **along** the leg the vessel has come. The switching test of §5-4 uses it.

y_e^p — how far **to the side**. The guidance law drives this one to zero.

Both are projections of the same vector onto the same orthonormal pair, so $x_e^2 + y_e^2 = \|\mathbf{p} - \mathbf{p}_i^n\|^2$ — a rotation preserves length, and that is the cheapest check on the whole construction.

Reading the figure.

Where to look	What is there	Why
the dashed pair at waypoint k	$\hat{\mathbf{t}}$ along the leg, $\hat{\mathbf{n}}$ to starboard	step 2: the leg carries its own frame
the black arrow	the position error, before it is resolved	the one vector both errors are components of
the brown segment on the leg	x_e — how far along	step 3, first row
the red segment perpendicular to it	y_e — how far off	step 3, second row
the right angle where they meet	$\hat{\mathbf{t}} \perp \hat{\mathbf{n}}$	which is what makes them independent
the grey arc at the waypoint	π_p , measured from north	step 1

Three properties worth stating before they are used.

1. **It is a rotation, so it preserves length.** $x_e^2 + y_e^2 = \|\mathbf{p} - \mathbf{p}_k\|^2$ exactly. That identity is the cheapest check on the whole construction, and it is what `verify_w05_guidance` check 5 tests.
2. **The sign of y_e is a convention with consequences.** Written as above, $y_e > 0$ means the vessel is to **starboard** of the leg, and §5-3's law then subtracts a positive correction — turning to port, back towards the path. Swap the two sine terms and the vessel runs away from the path at the same speed.
3. **Do not compute y_e from a distance formula.** The perpendicular distance $|(N - N_k) \sin \pi_p - (E - E_k) \cos \pi_p|$ gives the magnitude but throws away the sign and gives no x_e at all — and §5-4 needs x_e . One rotation answers both questions and cannot make them disagree.

! The same two lines, in Fossen's notation

This is the form used throughout the field, and the reference implementation is Fossen's TTK4190 lecture code — the source of the version taught in 2. USV 제어기 설계:

```
pi_p = atan2(yk_next-yk, xk_next-xk);          % path-tangential angle w.r.t. North
% along-track and cross-track errors (x_e, y_e) expressed in NED
x_e = (x-xk) * cos(pi_p) + (y-yk) * sin(pi_p);
y_e = -(x-xk) * sin(pi_p) + (y-yk) * cos(pi_p);
```

with (x, y) north and east. The **guidance** block of this week is these lines, and §5-4's switching test is the next two. The rotation gives the same y_e as MSS **crosstrackwpt.m** to machine precision (**verify_w05_guidance** check 5).

- The goal of path following is $y_e \rightarrow 0$; x_e is not driven anywhere, it is only **watched**, and §5-4 is what watches it.

Experiment 5-1 · The guidance layer on the canvas (10 min)

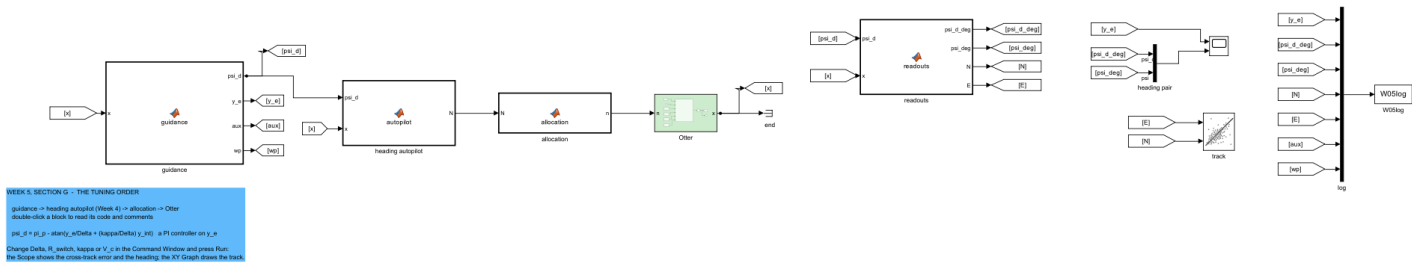
What it measures. Nothing yet: the four blocks of the table above are matched to the canvas, and the rotation that produces y_e is read inside the block that performs it.

The canvas. **W05_G_tuning** — the fullest model of the week, read here and measured in Experiment 5-6. Everything except the **guidance** block is Week 4, unchanged and not retuned.

Opening and running.

```
W05_0_setup
open_system('W05_G_tuning')          % Run - XY 그래프에 항적이 그려진다 / the track is drawn live
```

- Double-click **guidance** and read the numbered comments: the active leg, the rotation into (x_e, y_e) , the switching test of §5-4, and the law of §5-3 and §5-5.
- Double-click **heading autopilot** : the two gains of Week 4 and nothing else. The path does not appear anywhere in it.



In the figure	Meaning
guidance	the guidance law of §5-3 and §5-5; outputs ψ_d , y_e , the integral aux and the leg number wp
heading autopilot	the Week 4 gains, D on the yaw rate (§5-1), not retuned
allocation , Otter	the thrust split of Week 4 and the vessel
Goto [x] after the Otter	the twelve states, sent to the guidance, the autopilot and the readouts
readouts	the headings in degrees and the position, for display only
Scope, track (XY Graph), W05log	the cross-track error and headings; the track, east against north; the log

In class

- **Purpose** — show where guidance sits: one block in front of the autopilot of Week 4, which is not touched.
- **Point to** — double-click **guidance** and read the numbered comments: the leg, the rotation to y_e , the switching test, the law.
- **Ask** — "What does the autopilot know about the path?" Nothing. It follows ψ_d ; the path lives only in the guidance.
- **Take away** — guidance is a controller whose output is another controller's command.

5-2. Aiming at a point is not following a line

This section answers: why not simply point at the next waypoint?

What is observed. The vessel starts 20 m east of a path running north and is steered by the obvious law, $\psi_d = \text{atan2}(E_{k+1} - E, N_{k+1} - N)$ — point at the next waypoint:

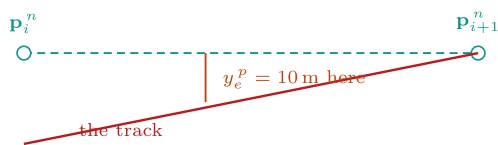
- halfway to the waypoint it is still **10.75** m off the path, more than half its starting offset,
- it reaches the path only **at** the waypoint, and then leaves it again on the next leg,
- while the law of §5-3, given the same start, is within **0.04** m of the path at the same point.

What follows from it, one line at a time.

1. **atan2** closes the distance to a **point**. The line it draws is the line from the vessel to the waypoint, and that line is not the path unless the vessel is already on it.
2. The quantity that measures "off the path" is therefore not the distance to the waypoint but the perpendicular distance to the **line**, the cross-track error y_e of §5-1.
3. **atan2** never uses y_e , so nothing in it drives y_e to zero: the error falls only as a side effect of arriving, which is why it vanishes exactly at the waypoint and not before.
4. A guidance law that follows a path must take y_e as its input and return a heading that reduces it. That is §5-3, and it makes the law a feedback controller like every other one in this course.
5. On a real mission the waypoints are far apart and the space **between** them is where the vessel is asked to be, so the error of line 1 is the whole mission rather than an approach transient.

(a) **atan2** — aim at the waypoint

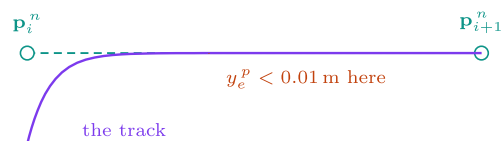
$$\psi_d = \text{atan2}(y_{i+1}^n - E, x_{i+1}^n - N)$$



The law regulates the distance to a **point**. The line that point sits on plays no part in it, so the vessel closes on the waypoint along a straight line of its own and is on the path only at the very end — if at all.

(b) **LOS** — aim Δ ahead on the path

$$\psi_d = \pi_p - \arctan(y_e^p / \Delta), \quad \Delta = 5 \text{ m}$$



The law regulates the distance to the **line**. The aim point moves along the path as the vessel does, so the correction fades to zero exactly when the vessel reaches the path, and the rest of the leg is run along it.

Both vessels start 20 m off the path and both reach the waypoint. Only one travels along the line to it. The tracks are not sketched: the LOS curve is the guidance law integrated along the path, $dy_e^p/ds = -y_e^p / \sqrt{\Delta^2 + y_e^2}$, by `_tools/w05_track_curves.awk`, so the shape is the law's and not a sketch. The drawn curves are that ideal kinematics; Experiment 5-2 runs the same start through the autopilot and the hull and measures 10.75 m against -0.04 m at the halfway point.

Reading the figure.

Where to look	What is there	Which line predicts it
(a), the red track	a straight line from the start to the waypoint	line 1: <code>atan2</code> closes the distance to a point , and the shortest way to a point is a straight line
(a), the label at halfway	$y_e = 10$ m, half the starting offset, purely by geometry	line 3: nothing in the law is acting on y_e
(a), where the track meets the path	only at the waypoint itself	line 3: the error vanishes as a side effect of arriving
(b), the purple track	it bends onto the path and then runs along it	line 4: the law takes y_e as its input
(b), the label at halfway	under 0.01 m — the law has already finished	§5-3: the correction fades as the error does
both panels	the same start, the same waypoint, the same vessel	the only difference is which quantity the law reads

- The drawn curves are the guidance law **integrated along the path**, $dy_e/ds = -y_e/\sqrt{\Delta^2 + y_e^2}$, not sketches — so the shape is the law's own. Experiment 5-2 puts the same start through the autopilot and the hull, where the vessel lags slightly:

Measured by Experiment 5-2, below:

Law	y_e at 50 m north	at 100 m (the waypoint)	at 150 m
atan2	10.75 m	0.50 m	-0.74 m
LOS (§5-3)	-0.04 m	0.01 m	0.01 m

Experiment 5-2 · Aiming at the waypoint (10 min)

What it measures. Lines 1 and 3: how far from the path the obvious law leaves the vessel, at three points along one leg, against the law that uses y_e .

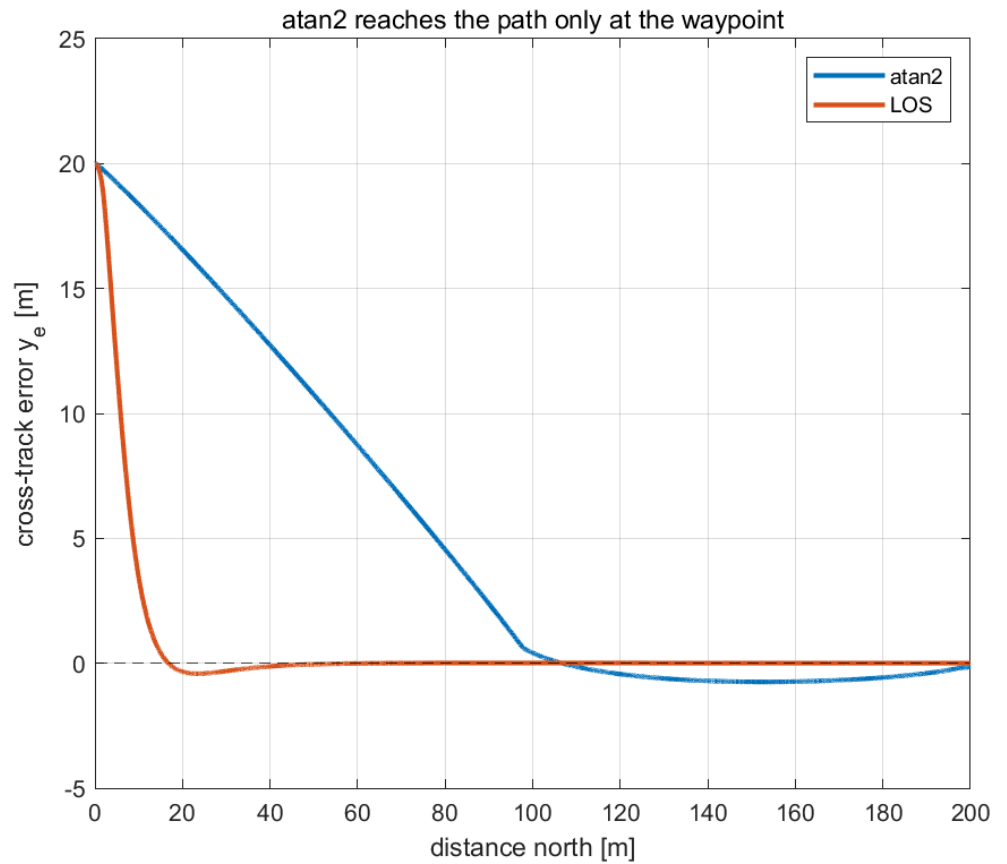
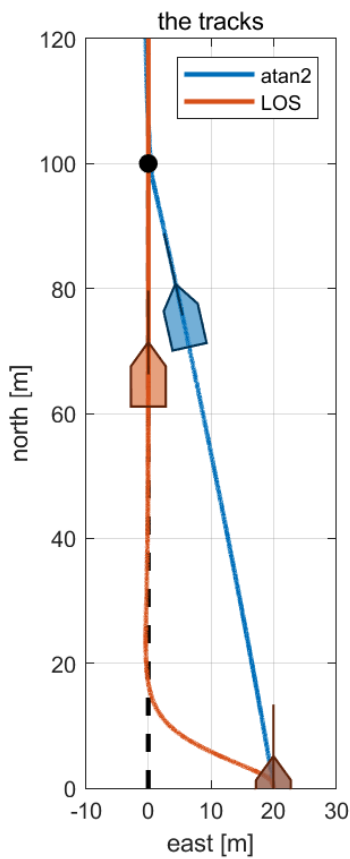
The model. `W05_C_atan2` — the guidance block switched to the `atan2` law, everything behind it (autopilot, allocation, Otter) exactly as Week 4 left it. The waypoints are set to one straight path north, so that the two laws differ in nothing but their own definition.

Opening and running.

```
W05_0_setup
WP_N = [0 100 200 300]'; WP_E = [0 0 0 0]';    % 북쪽으로 곧은 경로 / a straight path north
open_system('W05_C_atan2')                      % Run — XY 그래프에서 대각선으로 접근한다 / a diagonal approach
W05_C_aim_at_the_waypoint                        % 두 법칙을 나란히 / both laws, side by side
```

Expected output:

```
W05 Experiment 5-2 aiming at the waypoint against line of sight (start 20 m east of the path)
law      y_e at N = 50 m    N = 100 m    N = 150 m
atan2    10.75             0.50       -0.74
LOS      -0.04             0.01       0.01
```



In the figure	Meaning
left	the two tracks with hull silhouettes; the dashed line is the path, the dot the waypoint at 100 m
right	the cross-track error against the distance north

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
left panel, the straightness of the atan2 track	a diagonal drawn at the waypoint	line 1: the law aims at a point, and a constant bearing gives a straight run
right panel, the atan2 curve at $N = 100$ m	it touches zero, and only there	line 3: the error vanishes on arrival, not before
right panel, the atan2 curve past the waypoint	it leaves zero again, -0.74 m	line 1 on the next leg: a new point, a new diagonal
right panel, the LOS curve	on zero within about 20 m and staying there	line 4: a law that is fed y_e drives y_e
left panel, the hull silhouettes on the atan2 track	pointing at the waypoint throughout	line 1 again: that heading is exactly what the law asks for

What the figure says

- atan2 draws a straight line to the waypoint and meets the path only there; LOS is on the path within 20 m.

What to try	What to watch
<code>E0 = 5;</code> Run	2.80 m left at $N = 50$ m instead of 10.75 — the same fraction of the starting offset, 0.56 against 0.54. Line 1 is a geometric fault, not a matter of size; LOS leaves 0.02 m in the same run
<code>WP_N = [0 400]'; WP_E = [0 0]'; E0 = 20;</code> Run	on one long leg the atan2 track is 10.70 m off at the halfway point and 5.58 m off three-quarters of the way: line 5, the error is the mission

In class

- **Purpose** — show that "go to the waypoint" and "follow the path" are different goals.
- **Point to** — the blue track cutting diagonally to the waypoint while the orange one turns onto the path at once.
- **Ask** — "Why is atan2 still 10.75 m off halfway?" It reduces the distance to the waypoint, and the line to the waypoint is not the path.
- **Take away** — guidance must measure the distance to the line, y_e , and act on it.

5-3. Line of sight: a P controller on the cross-track error

This section answers: what law brings the vessel onto the line, and what is its one tuning knob?

The law. Instead of the waypoint, aim at a point on the path a distance Δ ahead:

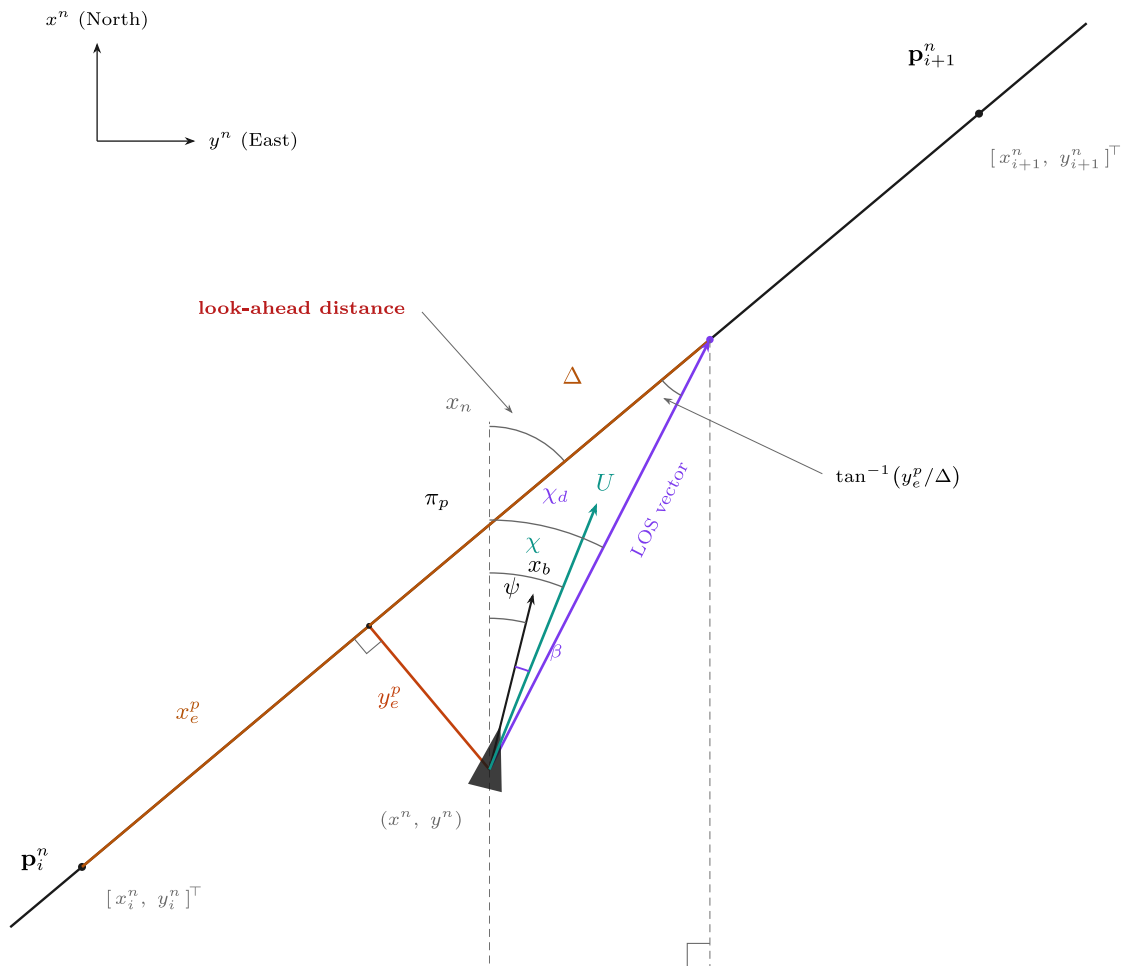
$$\psi_d = \pi_p - \arctan\left(\frac{y_e}{\Delta}\right)$$

- π_p says "line up with the path"; the arctan says "lean towards it, by an angle that grows with how far off the vessel is". On the path, $y_e = 0$ and the vessel simply heads along the leg; far from it, the lean approaches 90° and the vessel heads straight at the path.

Where the arctan comes from. It is not a choice of function — it is the angle of a right triangle. Drop a perpendicular from the vessel to the path; its foot is the point on the path abeam the vessel. Now step a distance Δ along the path from that foot: that is the aim point. The triangle with legs Δ (along the path) and y_e (across it) has

$$\tan(\text{the lean}) = \frac{y_e}{\Delta} \quad \Rightarrow \quad \text{the lean} = \arctan\left(\frac{y_e}{\Delta}\right)$$

and the command is π_p turned by that lean, towards the path — hence the minus sign, with y_e positive to starboard.



$$\pi_p = \text{atan2}(y_{i+1}^n - y_i^n, x_{i+1}^n - x_i^n)$$

Known

$$y_e^p = -(x^n - x_i^n) \sin \pi_p + (y^n - y_i^n) \cos \pi_p$$

Known

$$\mathbf{p}_i^n = [x_i^n, y_i^n]^\top, \quad \mathbf{p}_{i+1}^n = [x_{i+1}^n, y_{i+1}^n]^\top$$

Known

$$\chi_d = \pi_p - \tan^{-1}(K_p y_e^p), \quad K_p = \frac{1}{\Delta}$$

Want to compute

$$y_e^p \rightarrow 0, \quad \chi_d \rightarrow \pi_p$$

On the path the correction vanishes and the vessel steers straight along the leg. That is the whole objective, written as a limit.

Two terms, and that is all. π_p lines the vessel up *with* the path; $-\tan^{-1}(y_e^p/\Delta)$ leans it *towards* the path. Nothing else appears, which is exactly why §5-7 can say what the law cannot do.

The correction is bounded by 90° . As $y_e^p \rightarrow 0$ the vessel is on the path and $\chi_d \rightarrow \pi_p$; as $y_e^p \rightarrow \infty$ the law gives $\chi_d \rightarrow \pi_p - 90^\circ$, which heads straight at the path and never away from it.

Δ is measured along the path, from the foot of the perpendicular and not from the vessel. The vessel's own distance to the aim point is $\sqrt{\Delta^2 + (y_e^p)^2}$, which is larger.

Notation, and the numbers drawn here

Waypoints are $\mathbf{p}_i^n = [x_i^n, y_i^n]^\top$, resolved in $\{n\}$. The two errors carry the superscript p because they are resolved in the **path** frame $\{p\}$ — not in $\{n\}$ and not in $\{b\}$. The superscripts are what keep the three frames apart.

Drawn with $\pi_p = 50^\circ$, $x_e^p = 32$ m, $y_e^p = 16$ m and $\Delta = 38$ m, so the correction is $\tan^{-1}(16/38) = 22.83^\circ$ and $\chi_d = 27.17^\circ$; the line-of-sight distance is 41.23 m. The hull is at $\psi = 14^\circ$ with a crab angle $\beta = 8^\circ$, so $\chi = 22^\circ$ — not yet on the commanded course, which is what makes the three arcs at the hull distinct.

Course or heading. The law as written commands a *course* χ_d . This course carries a heading autopilot from Week 4, so §5-3 commands ψ_d : the same expression with ψ_d in place of χ_d . The two agree exactly when $\beta = 0$; Week 4 §4-4 defines the gap and §5-5 measures what it costs on a path.

Reading the figure.

Where to look	What is there	Why
the right angle on the path	the foot of the perpendicular, abeam the vessel	it is the origin the look-ahead is measured from
the segment marked Δ	measured along the path , from the foot	not from the vessel: the vessel's own distance to the aim point is $\sqrt{\Delta^2 + y_e^2}$, which is longer
the purple ray	the line of sight, vessel to aim point	the name of the law: the vessel looks at a point and steers at it
the arc marked $\arctan(y_e/\Delta)$	the lean, between the path direction and the line of sight	the right triangle above
π_p at the lower waypoint	the path direction, from north	§5-1 step 1
the two brown segments	x_e and y_e of §5-1	the law uses only the second of them
the three arcs at the hull	ψ , χ and χ_d — heading, course and commanded course	they differ by the crab angle of Week 4 §4-4, which is §5-5's subject

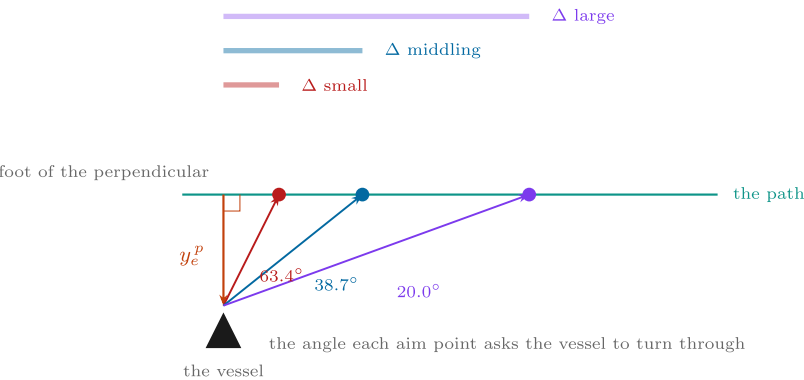
- **Two terms, and that is all.** π_p lines the vessel up *with* the path; — $\arctan(y_e/\Delta)$ leans it *towards* the path. Nothing else appears — which is exactly why §5-5 can say what this law cannot do.
- **The correction is bounded by 90° .** As $y_e \rightarrow 0$ the lean goes to zero and $\psi_d \rightarrow \pi_p$; as $y_e \rightarrow \pm\infty$ it approaches $\mp 90^\circ$, heading straight at the path and never away from it. A law built from a raw gain $-Ky_e$ has no such bound and would command absurd headings far from the path.

It is a P controller. For errors small against Δ , $\arctan(y_e/\Delta) \approx y_e/\Delta$ (within 1.4 % for $|y_e| \leq 0.2\Delta$, `verify_w05_guidance` check 2), and

$$\psi_d - \pi_p \approx -\frac{1}{\Delta} y_e \quad \implies \quad K_p = \frac{1}{\Delta}$$

Symbol	Quantity	Value
Δ	look-ahead distance	5 m, <code>w05_0_setup.m</code>
$1/\Delta$	the P gain: heading correction per metre of error	0.2 rad/m

So Δ is tuned like K_p in Weeks 2 to 4, backwards: a **smaller Δ** is a **larger** gain.



Reading the figure.

the trade

One and the same y_e^p . Only Δ has changed, and the command has changed by a factor of three.

Δ too small — the arctan saturates almost at once, the vessel turns nearly perpendicular to the path and arrives with speed across it. Measured at $\Delta = 0.5$ m: 1.35 m of overshoot and seven crossings of the path.

Δ too large — the correction is gentle and the vessel closes slowly. At $\Delta = 40$ m it needs 118.3 s to come within 1 m of the path, against 32.4 s at $\Delta = 5$ m.

The rule of thumb is $\Delta = 2$ to 5 hull lengths; for the 2 m Otter that is 4 to 10 m, and this course uses 5 m — two to three lengths. **Experiment 5-3 is the evidence, and the rule of thumb only the sanity check.**

A third cost appears in §5-5: the steady offset a current leaves is $\Delta \tan \beta$, **linear in Δ** .

Where to look	What is there	Why
the three aim points on the path	all three lie on the path, at Δ = small, middling, large from the same foot	only Δ has changed
the single red segment y_e	one and the same error in all three cases	so the figure isolates Δ and nothing else
the three angles, 53.1° , 38.7° , 20.0°	what each aim point asks the vessel to turn through	$\arctan(y_e/\Delta)$ with one numerator and three denominators
the ratio between them	a factor of about three in the command, from a factor of three in Δ	Δ is the gain, read backwards

- The trade is the P sweep of Week 2 §2-6 in new clothing: too small a Δ saturates the arctan, turns the vessel nearly perpendicular to the path, and arrives with speed across it; too large a Δ closes gently and slowly.

Measured by Experiment 5-3, below:

Δ [m]	$1/\Delta$	within 1 m after [s]	overshoot [m]	zero crossings
0.5	2.000	27.7	1.35	7
1	1.000	27.9	0.70	2
2.5	0.400	29.1	0.42	2
5	0.200	32.4	0.41	2
10	0.100	41.7	0.41	2
20	0.050	65.2	0.40	1
40	0.025	118.3	0.40	1

- A large Δ is slow; a small one overshoots and swings. $\Delta = 5$ m — two to three hull lengths — reaches the path almost as fast as the smallest values without swinging, and is the choice.

Experiment 5-3 · The look-ahead distance (15 min)

What it measures. That Δ behaves exactly as $1/K_p$: the sweep below is the P sweep of Week 2 §2-6 read backwards, on one long leg so that nothing but the gain is in play.

The model. **W05_D_LOS** — the guidance block switched to the LOS law. The waypoints are one 400 m leg, long enough for the transient to finish before any corner.

Opening and running.

```

W05_0_setup
WP_N = [0 400]'; WP_E = [0 0]';    % 긴 다리 하나 / one long leg
open_system('W05_D_LOS')           % Run - Delta = 5 m / the chosen value
Delta = 0.5;                        % Run - 지그재그로 다가간다 / it zigzags onto the path
Delta = 40;                         % Run - 한참 걸린다 / it creeps
W05_D_lookahead_distance            % 일곱 값을 한 번에 / all seven values at once

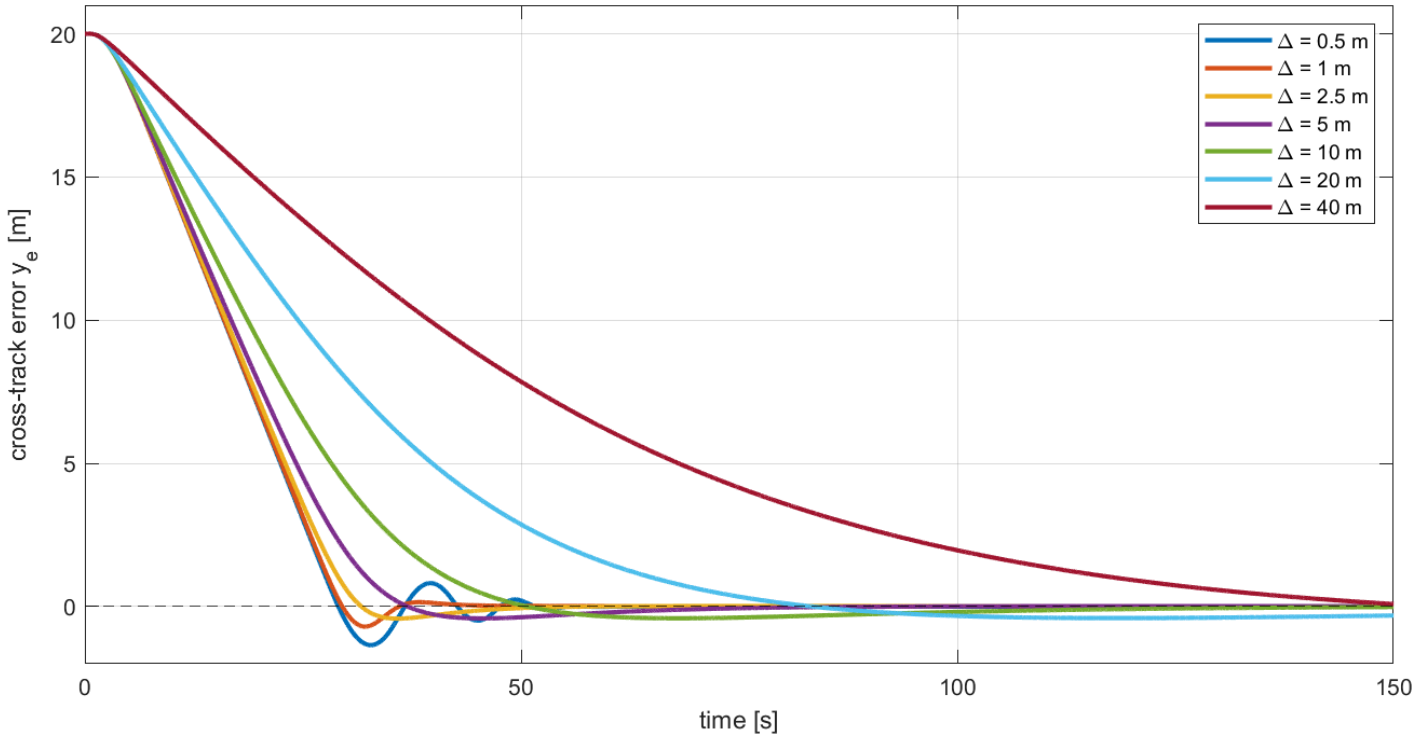
```

Expected output:

W05 Experiment 5-3 LOS, the look-ahead distance (start 20 m east of the path)

Delta [m]	P gain 1/Delta	within 1 m after [s]	overshoot [m]	zero crossings
0.5	2.000	27.7	1.35	7
1	1.000	27.9	0.70	2
2.5	0.400	29.1	0.42	2
5	0.200	32.4	0.41	2
10	0.100	41.7	0.41	2
20	0.050	65.2	0.40	1
40	0.025	118.3	0.40	1

LOS as a P controller: a smaller Δ is faster and overshoots more



In the figure	Meaning
traces	the cross-track error against time from 20 m off the path, one per Δ

Reading the figure against the law.

Where to look	What is there	What it corresponds to
the steepness of each trace at the start	steeper the smaller Δ is	$K_p = 1/\Delta$: a larger gain pushes harder on the same error
the $\Delta = 0.5$ m trace crossing zero repeatedly	7 crossings, overshoot 1.35 m	too large a gain rings, as in Week 2 §2-6 — here the ringing is a zigzag across the path
the $\Delta = 40$ m trace	within 1 m only after 118.3 s	too small a gain is slow, and $1/40$ is a very small gain
the middle traces, $\Delta = 2.5$ to 10 m	all reach 1 m within 29 to 42 s with the same 0.41 m of overshoot	the flat part of the trade-off, where the choice is made
every trace, once on the path	it stays there	the law has no steady-state error to leave on a straight path with no current — §5-5 is where that changes

What the figure says

- The curves fan out exactly as a P gain sweep does: the small Δ drops fast and crosses back and forth; the large one creeps.

What to try	What to watch
Delta = 0.2; Run	the zigzag never stops: 44 crossings of the path and still 1.20 m off at the end of the leg. This is a K_p raised past its limit, with the hull's own turn rate as the slow element
Delta = 100; Run	one crossing and no zigzag at all, but the path is reached only after 286 s — a gain of 0.01 rad/m spends the whole leg converging

In class

- **Purpose** — recognise a familiar controller in an unfamiliar law.
- **Point to** — the $\Delta = 0.5$ m trace crossing zero repeatedly; the $\Delta = 40$ m trace still 5 m off after 100 s.
- **Ask** — "Which way does the gain go when Δ is halved?" Up: $K_p = 1/\Delta$ doubles.
- **Take away** — tune Δ as K_p was tuned: the smallest value that does not swing.

5-4. When to move to the next leg

This section answers: at a corner, when does guidance switch to the next leg?

What is observed. The same mission is run with five switching distances, and the corners are where the tracks differ:

- with $R = 20$ m the vessel turns early and passes almost **20** m from the new leg,
- with $R = 1$ m it runs **past** the corner before turning, and sweeps **3.70** m outside the sharpest one,
- and the mission finishes sooner the larger R is: **305** s against **364** s.

What follows from it, one line at a time.

1. The switch is a test on the **along-track** coordinate: the guidance moves to the next leg when less than R of the active leg remains, $d - x_e < R$, with d the leg's length. This is the test MSS uses, and it is a distance along the path rather than a circle round the waypoint.

The two tests, and why they are not the same. Both are one line of code and they disagree:

along-track: $d - x_e < R$

acceptance circle: $\|\mathbf{p} - \mathbf{p}_{k+1}\| \leq R$

Symbol	Quantity	Value · source
d	the length of the active leg	m · $\ \mathbf{p}_{k+1} - \mathbf{p}_k\ $
x_e	how far along it the vessel has come	m · §5-1
$d - x_e$	what is left to run along the leg	m · negative once the vessel is past the waypoint
R	the switching distance	3 m · <code>R_switch</code> , chosen in line 4

Writing the true distance in path-frame components shows the relation exactly:

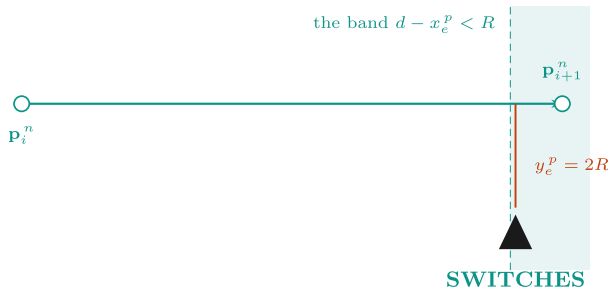
$$\|\mathbf{p} - \mathbf{p}_{k+1}\| = \sqrt{(d - x_e)^2 + y_e^2} \geq d - x_e$$

2. **On the path the two agree; off it they never do.** With $y_e = 0$ the square root collapses to $d - x_e$ and the tests are identical. With $y_e \neq 0$ the circle test is **always the stricter of the two**, by an amount that grows with how far off the path the vessel is — so a vessel that is running wide can satisfy the along-track test and fail the circle at the same instant.
3. **The circle can be missed altogether.** A vessel whose cross-track error never falls below R never enters the circle, passes the waypoint, and waits for an arrival that has already happened — for ever. The along-track test cannot fail that way: $d - x_e$ goes negative the moment the vessel crosses the line through the waypoint perpendicular to

the leg, whatever its cross-track error. This is not a hypothetical; Week 9 §9-5 loses a whole mission to it at **1.3** m/s of current.

(a) along-track — what MSS uses

(b) circle of acceptance — what most books draw



The test looks only at how much of the leg is left, so being far off the path cannot defeat it: a vessel blown wide still moves on to the next leg.

if ($d - x_e < R_{\text{switch}}$) $k = k + 1$



The test is a true distance, so a vessel that never enters the circle waits for a waypoint it has already passed — and can wait for ever.

if ($\text{norm}(\mathbf{p} - \mathbf{wp_next}) < R$) $k = k + 1$

On the path the two agree exactly, because $y_e^p = 0$ makes $\|\mathbf{p} - \mathbf{p}_{i+1}^n\| = d - x_e^p$. Off it they do not: $\|\mathbf{p} - \mathbf{p}_{i+1}^n\| = \sqrt{(d - x_e^p)^2 + y_e^2} \geq d - x_e^p$, so **the circle test is always the stricter of the two**, by an amount that grows with y_e^p . Experiment 5-4 evaluates both on one position: the along-track remainder is 4.50 m and passes; the true distance is 10.97 m and fails. The one hard limit is $R < \min_k d_k$ — otherwise a leg is already “reached” before the vessel has travelled any of it.

Reading the figure.

Where to look	What is there	Which line predicts it
(a), the shaded band before the waypoint	the region where $d - x_e < R$	line 1: the test is a distance along the leg
(a), the vessel a whole $y_e = 2R$ off the path	still inside the band, and it switches	line 3: being off the path cannot defeat the along-track test
(b), the circle of radius R round the waypoint	the region where $\ \mathbf{p} - \mathbf{p}_{k+1}\ \leq R$	the other test, drawn
(b), the same vessel, outside the circle	10.97 m away, and it does not switch	line 2: the circle is the stricter test
the two code lines under the panels	$d - x_e < R_{\text{switch}}$ against $\text{norm}(\mathbf{p} - \mathbf{wp_next}) < R$	one line of difference, and the whole behaviour
the caption's arithmetic	along-track remainder 4.50 m passes; true distance 10.97 m fails	line 2: $\sqrt{(d - x_e)^2 + y_e^2} \geq d - x_e$, here by 6.47 m

- There is one hard limit on R** , and it belongs to the along-track test: $R < \min_k d_k$. If R exceeds a leg's own length, then $d - x_e < R$ is true the instant the leg becomes active, and that leg is "reached" before the vessel has travelled any of it.
- A large R cuts the corner.** At the instant of the switch the vessel is still on the old leg, R before its end, so it is about R away from the new leg — and the new leg is what y_e is now measured against. Hence the largest distance is close to R itself in the last three rows.
- A small R carries the vessel past the corner.** The turn begins only at the corner, and the hull cannot turn instantly: at **0.77** m/s it sweeps outside, **3.70** m at the **135°** corner for $R = 1$ m. The sharper the corner, the further it goes.
- Lines 5 and 6 pull in opposite directions, so the best R is the one whose **worst** corner is smallest — $R = 3$ m, at **2.98** m.
- Cutting corners is not wrong in itself; it also finishes sooner. The choice made here is to stay close to the path, and a survey that must cover its lines exactly would choose differently.

Measured by Experiment 5-4, below, on the five-waypoint mission with $\Delta = 5$ m:

<i>R</i> [m]	largest distance from the new leg at corners 1, 2, 3 [m]	last waypoint reached at [s]
1	2.06, 2.08, 3.70	363.5
3	2.98, 2.98, 2.29	352.8
5	4.98, 4.98, 3.53	344.0
10	9.97, 9.98, 7.04	328.0
20	19.97, 19.98, 14.08	305.4

Experiment 5-4 · Waypoint switching (10 min)

What it measures. Lines 5, 6 and 7: how far from the new leg each switching distance leaves the vessel at the three corners, and how long the mission then takes.

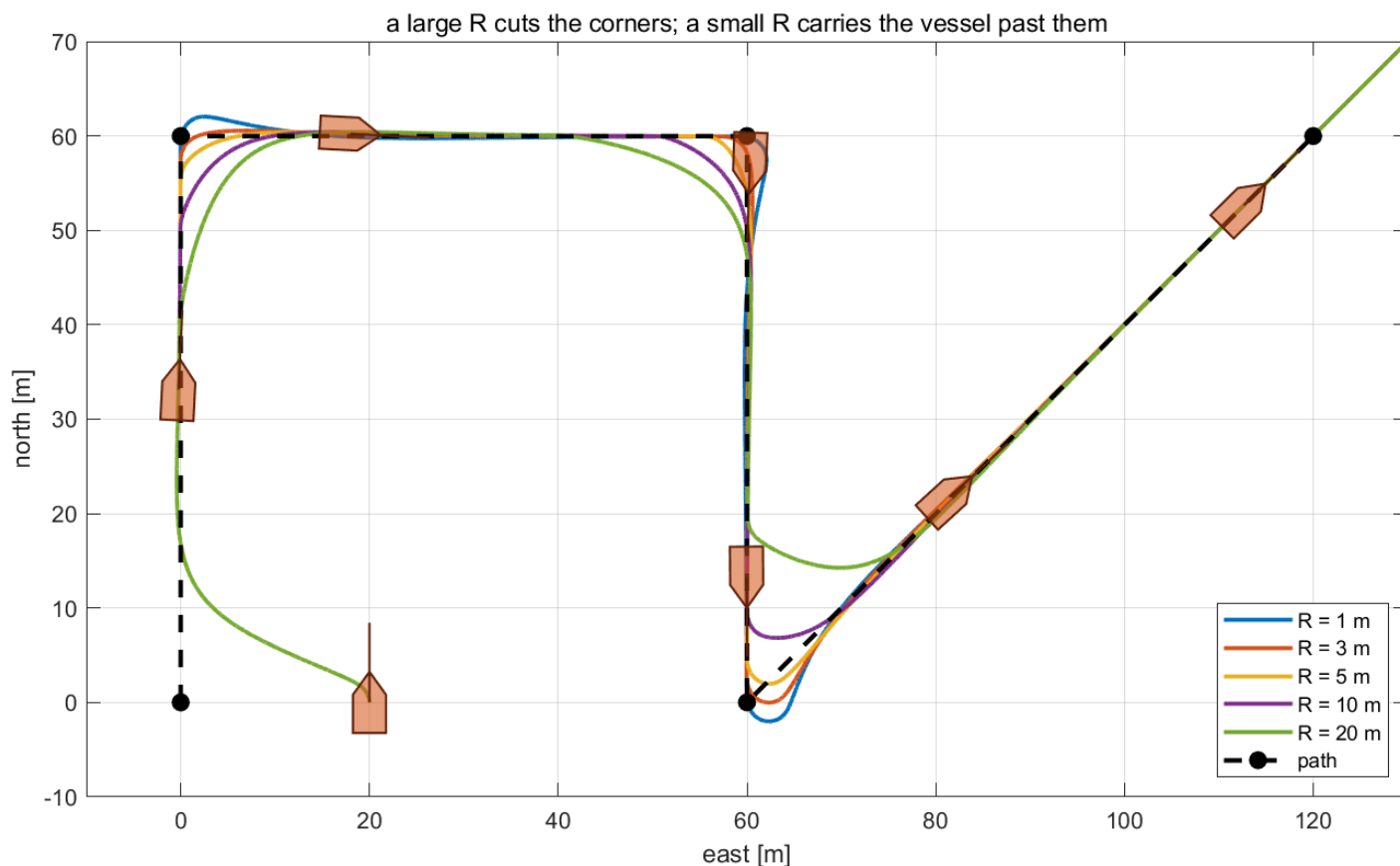
The model. `W05_E_switching` — the LOS law on the full five-waypoint mission, with Δ fixed at the value Experiment 5-3 chose. Only `R_switch` changes between runs, and the XY Graph shows the corners as they are rounded.

Opening and running.

```
W05_0_setup
open_system('W05_E_switching')      % Run - R = 3 m / the chosen value
R_switch = 20;                      % Run - 모퉁이를 크게 자른다 / it cuts the corners wide
R_switch = 1;                       % Run - 모퉁이를 지나쳐서 돈다 / it runs past them
W05_E_waypoint_switching            % 다섯 값을 한 번에 / all five values at once
```

Expected output:

```
W05 Experiment 5-4  waypoint switching on the mission  (LOS, Delta = 5 m, no current)
  R_switch [m]    largest distance from the new leg at corners 1, 2, 3 [m]    last waypoint at [s]
    1              2.06    2.08    3.70                                363.5
    3              2.98    2.98    2.29                                352.8
    5              4.98    4.98    3.53                                344.0
   10              9.97    9.98    7.04                                328.0
   20             19.97   19.98   14.08                                305.4
```



In the figure	Meaning
coloured tracks	the mission for $R = 1$ to 20 m; hulls drawn on $R = 3$ m
dashed line and dots	the path and its five waypoints

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
the $R = 20$ m track at the first corner	it leaves the leg some 20 m early	line 2: the switch happens R before the end of the leg
the largest-distance columns for $R = 5, 10, 20$	4.98, 9.97, 19.97 m — each equal to R	line 2 again, in numbers
the $R = 1$ m track at the third corner	the widest sweep of all, 3.70 m	line 3: the turn starts at the corner, and the 135° corner is the sharpest
the $R = 3$ m columns	2.98, 2.98, 2.29 m — the smallest worst case	line 4: the compromise between the two
the last column, top to bottom	363.5 → 305.4 s	line 5: cutting corners shortens the mission
the straight parts between corners	every track lies on the path	§5-3: the switching distance decides the corners only

What the figure says

- The green track rounds every corner early and wide; the blue one runs past each corner before turning. $R = 3$ m sits between them.

What to try	What to watch
<code>R_switch = 50;</code> Run	almost as long as a 60 m leg: the guidance switches as soon as a leg begins, the vessel is never nearer than 49.98 m to the leg it is supposed to follow, and it reaches the last leg in 146.9 s by flying the diagonal
<code>Delta = 20;</code> <code>R_switch = 3;</code> Run	the worst corner grows to 7.93 m although R has not changed — a gain of 1/20 cannot pull the vessel onto the new leg quickly, so Δ and R are read together

In class

- **Purpose** — the second tuning knob, set from what the corners look like.
- **Point to** — the corner at (60, 0), where the tracks spread the most.
- **Ask** — "Why does the largest distance equal **R** for large **R**?" At the switch the vessel is on the old leg, **R** before the corner — **R** from the new leg.
- **Take away** — too early cuts, too late overshoots; choose from the worst corner.

5-5. A current, and the integral that removes its offset (ILOS)

This section answers: what does LOS do when a current pushes the vessel sideways, and how is it fixed?

What is observed. The same LOS law, the same **Δ**, with a current flowing east across a path running north:

- the vessel settles **parallel to the path but beside it** — **2.107** m off in a **0.3** m/s current, and it stays there,
- while pointing **22.8°** into the current,
- and the offset grows with the current: **0.658, 1.344, 2.107, 3.012** m.

What follows from it, one line at a time.

1. To travel north across an eastward current the vessel must **point** west of north: the crab angle of Week 4 §4-4. Its course is along the path while its heading is not.
2. LOS produces that heading only through $\arctan(y_e/\Delta)$, so it can hold a heading away from π_p **only while** $y_e \neq 0$. This is Week 2 §2-6 line 7 once more: a proportional law makes its output out of error alone.
3. In the steady state the heading held equals the heading asked for, $\psi = \psi_d = \pi_p - \arctan(y_e/\Delta)$. Solving for the error,

$$y_e = \Delta \tan(\pi_p - \psi)$$

4. The offset is therefore **predictable, not mysterious**: it is **Δ** times the tangent of the crab angle the current forces. The measurements below match it to the millimetre (`verify_w05_guidance` check 3).
5. Reducing **Δ** would shrink the offset, by line 4 — but **Δ** was already chosen in §5-3 for the transient, and a smaller one zigzags. Two requirements, one knob: the standing answer of this course is to add the **integral** instead.

Measured by Experiment 5-5a, below, a current flowing east across a path north:

current [m/s]	y_e left [m]	heading held [deg]	$\Delta \tan(\text{heading})$ [m]
0.1	0.658	−7.5	0.658
0.2	1.344	−15.0	1.344
0.3	2.107	−22.8	2.107
0.4	3.012	−31.1	3.012

- The last two columns are line 4: prediction and measurement agree to the millimetre in every row. Heading is not course — the vessel points **22.8°** into a **0.3** m/s current and travels due north.

ILOS adds the integral. As in Weeks 2 to 4, the cure for the error a P controller leaves is an integral:

$$\psi_d = \pi_p - \arctan\left(\frac{y_e}{\Delta} + \frac{\kappa}{\Delta} y_{\text{int}}\right), \quad \dot{y}_{\text{int}} = \frac{\Delta y_e}{\Delta^2 + (y_e + \kappa y_{\text{int}})^2}$$

Symbol	Quantity	Value
y_{int}	the integral state	m·s scaled; starts at 0
κ	integral constant; the I gain is κ/Δ	0.3 , <code>w05_0_setup.m</code>

- Near the path $\arctan x \approx x$ again, so the law reads $\psi_d - \pi_p \approx -(1/\Delta)y_e - (\kappa/\Delta)y_{\text{int}}$: **PI on the cross-track error**, with $K_p = 1/\Delta$ from §5-3 and $K_i = \kappa/\Delta$.
- The integral state therefore does for the current what the integral of Week 3 did for the drag: it holds the crab angle by itself, so that line 2 no longer needs an error to produce one.
- The denominator $\Delta^2 + (y_e + \kappa y_{\text{int}})^2$ grows with the error, so the state barely integrates while the vessel is far off the path — **anti-windup built into the law**, in place of the back-calculation bolted on in Weeks 2 to 4 (Børhaug, Pavlov and Pettersen, 2008; Fossen, *Handbook*, 2nd ed., §12.3).

Measured by Experiment 5-5b at 0.3 m/s, with $\Delta = 5$ m:

κ	mean y_e , last 100 s [m]	overshoot [m]
LOS ($\kappa = 0$)	2.107	0
0.1	0.009	0.03
0.3	0.011	1.79
1	-0.011	4.10
3	0.169	6.32

- Every κ removes the offset — line 7 — and a larger one overshoots more, which is the price of every integral since Week 2. On a single long leg $\kappa = 0.1$ is the cleanest; §5-6 chooses on the mission, where the legs are short and the integral has less time.

Experiment 5-5a · The offset a current leaves under LOS (7 min)

What it measures. Lines 3 and 4: the error LOS settles at in four currents, against $\Delta \tan$ of the heading it holds.

The model. `w05_F_LOS` — the LOS law of §5-3 on one straight 400 m leg, long enough for the offset to settle. Only the current changes between runs.

Opening and running.

```
w05_0_setup
WP_N = [0 400]'; WP_E = [0 0]'; V_c = 0.3;      % 동쪽으로 0.3 m/s / 0.3 m/s to the east
open_system('w05_F_LOS')                        % Run - 경로와 나란히, 2.1 m 옆으로 / parallel, 2.1 m beside it
V_c = 0.4;                                       % Run - 더 비스듬히 서고 더 벌어진다 / further off
w05_F_LOS_in_a_current                          % 조류 네 가지를 한 번에 / all four currents at once
```

Expected output:

```

W05 Experiment 5-5a  LOS in a current flowing east  (Delta = 5 m)
  V_c [m/s]    error left [m]    heading held [deg]    Delta*tan(heading) [m]
  0.1                0.658                -7.5                0.658
  0.2                1.344                -15.0               1.344
  0.3                2.107                -22.8               2.107
  0.4                3.012                -31.1               3.012

```

- The last two columns are line 4, to the millimetre in every row. The vessel points **22.8°** into a **0.3** m/s current and travels due north — Week 4 §4-4, now costing a path error.

Experiment 5-5b · The integral that removes it (8 min)

What it measures. Line 7: the same current with the integral state switched on, for four values of κ — and what each one costs in overshoot.

The model. `W05_F_ILOS` — the same guidance block with the integral state of the law above. The black line in the figure is the run of Experiment 5-5a, for comparison.

Opening and running.

```

W05_0_setup
WP_N = [0 400]'; WP_E = [0 0]'; V_c = 0.3;
open_system('W05_F_ILOS')           % Run - 같은 조류, 오차가 사라진다 / the same current, no offset
kappa = 1;                          % Run - 더 빨리 없애지만 더 넘어간다 / faster, and past it
W05_F_ILOS_removes_it                % kappa 네 가지를 한 번에 / all four values at once

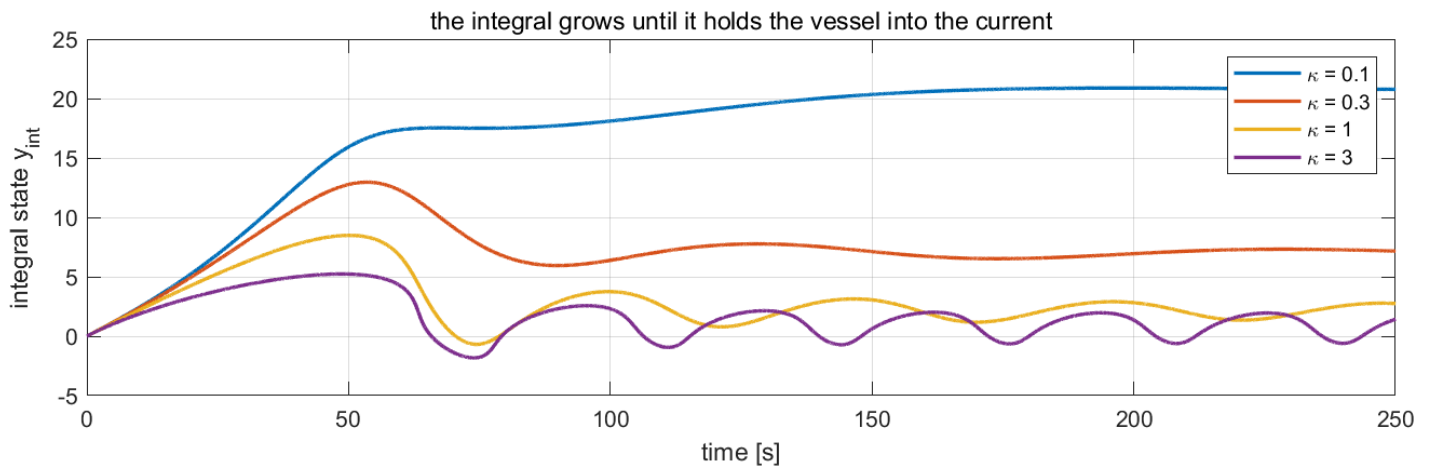
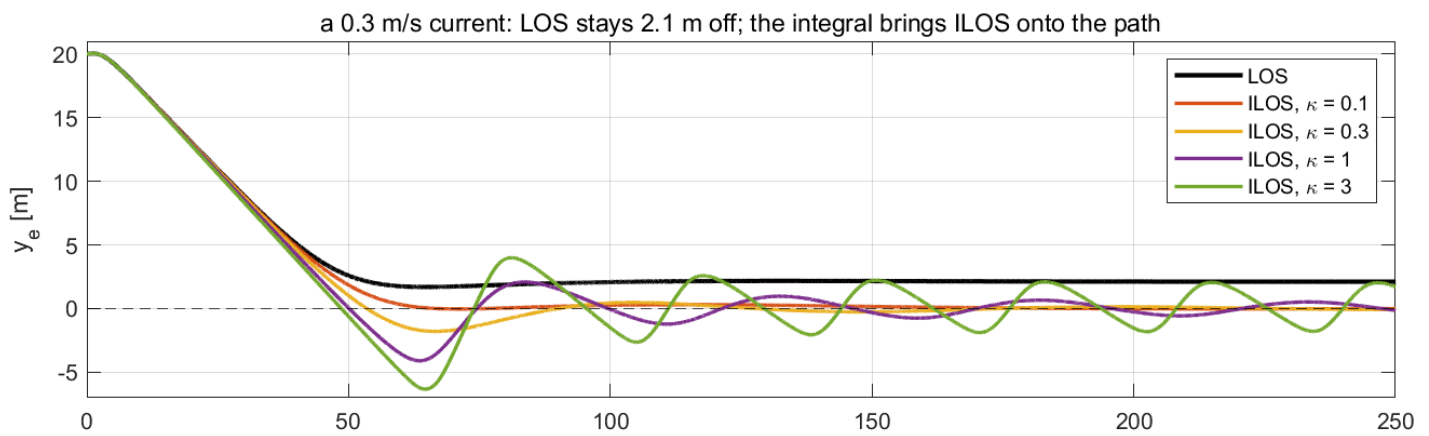
```

Expected output:

```

W05 Experiment 5-5b  ILOS at 0.3 m/s  (I gain = kappa / Delta)
  kappa    mean y_e, last 100 s [m]    overshoot [m]    integral state at 400 s
  0.1                0.009                0.03                20.95
  0.3                0.011                1.79                7.03
  1                -0.011                4.10                2.35
  3                0.169                6.32               -0.50

```



In the figure	Meaning
top	the cross-track error: LOS in black, ILOS for four values of κ
bottom	the integral state of each ILOS run

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
top panel, the black LOS line	flat at 2.107 m, for ever	lines 2 and 3: the error is what produces the lean, so it cannot go
the LOS table, last two columns	equal in all four rows	line 4: $y_e = \Delta \tan(\text{crab angle})$
top panel, every coloured trace	reaches zero and stays	line 7: the integral holds the lean instead
bottom panel, the states levelling off at 20.95, 7.03, 2.35	different values, same effect	line 6: what matters is the product $(\kappa/\Delta) y_{int}$, so a smaller κ needs a larger state
top panel, the overshoot at $\kappa = 1$ and 3	4.10 m and 6.32 m	the price of every integral since Week 2 §2-8: late correction arrives after it was needed
the first seconds of every trace, from 20 m off	no wind-up, no spike	line 8: while y_e is large the denominator is large and the state hardly moves

What the figure says

- LOS settles 2.1 m off the path; every ILOS settles on it, and a larger κ gets there by swinging past.

What to try	What to watch
$V_c = 0.5$; on <code>W05_F_LOS</code>	4.234 m off, holding -40.3° — and $5 \tan 40.3^\circ = 4.234$ m: line 4 holds at any current
$\Delta = 2.5$; $V_c = 0.3$; on <code>W05_F_LOS</code>	half the look-ahead distance halves the offset, 1.054 m at the same -22.9° heading, exactly as line 4 says. It also sharpens the transient, which is why §5-6 fixes Δ on the transient and leaves the current to κ

In class

- **Purpose** — the same story as the P and I of Week 2, now on a path.
- **Point to** — the black line holding 2.1 m; the integral states levelling off at different values that all produce the same steady lean into the current.
- **Ask** — "Why does LOS not reach the path, when its heading is exactly what it commanded?" To hold the line across the current, the vessel must point into it, and LOS only commands that while $y_e \neq 0$.
- **Take away** — a steady push leaves an error under P; the integral removes it.

5-6. The tuning order, applied to guidance

This section answers: in what order are the three parameters chosen?

The order of Week 2 §2-12: the proportional part first, the integral only for an error that remains.

Step	What is chosen	From what measurement	Choice
1	Δ , the P gain	Experiment 5-3: fast without swinging on a straight leg	5 m
2	R , the switching distance	Experiment 5-4: the smallest worst corner	3 m
3	κ , the I gain	Experiment 5-6: the mission in a 0.3 m/s current	below

Measured by Experiment 5-6, below, mean $|y_e|$ on each leg from 25 s after entering it:

κ	leg 1	leg 2	leg 3	leg 4	sum of legs 2–4 [m]	last waypoint at [s]
0 (LOS)	3.29	0.12	2.14	1.38	3.64	330.4
0.1	2.04	1.22	1.33	0.55	3.10	337.0
0.3	2.29	0.40	0.38	0.14	0.92	338.8
0.5	2.44	0.30	0.48	0.15	0.93	342.6
1	2.62	0.24	0.66	0.18	1.07	349.8

- Leg 1 includes the 20 m start offset and is left out of the comparison. LOS leaves 2.14 m and 1.38 m on the legs that cross the current and almost nothing on leg 2, which runs with it.
- On 60 m legs $\kappa = 0.1$ is too slow. $\kappa = 0.3$ and 0.5 are nearly equal; Experiment 5-5b showed a larger κ overshoots more, so $\kappa = 0.3$.

The result, $\Delta = 5$ m, $R = 3$ m, $\kappa = 0.3$, is the default in `W05_0_setup.m`. No model of the vessel was used; only what the Scope and the track showed.

i What is left out this week

The previous version of this week derived the laws in full, proved their stability with Lyapunov functions, and added a third law, ALOS, which estimates the crab angle directly. That material is kept in `W05_simulink/_previous_version/` for reference; nothing in this week's tuning depends on it.

Experiment 5-6 · The tuning order (15 min)

What it measures. Step 3 of the order above, on the mission the guidance is actually for: mean $|y_e|$ leg by leg, in a current, for five values of κ — and the time the mission takes.

The model. `W05_G_tuning` — ILOS on the five-waypoint mission with Δ and R already fixed by Experiments 5-3 and 5-4. Only κ changes.

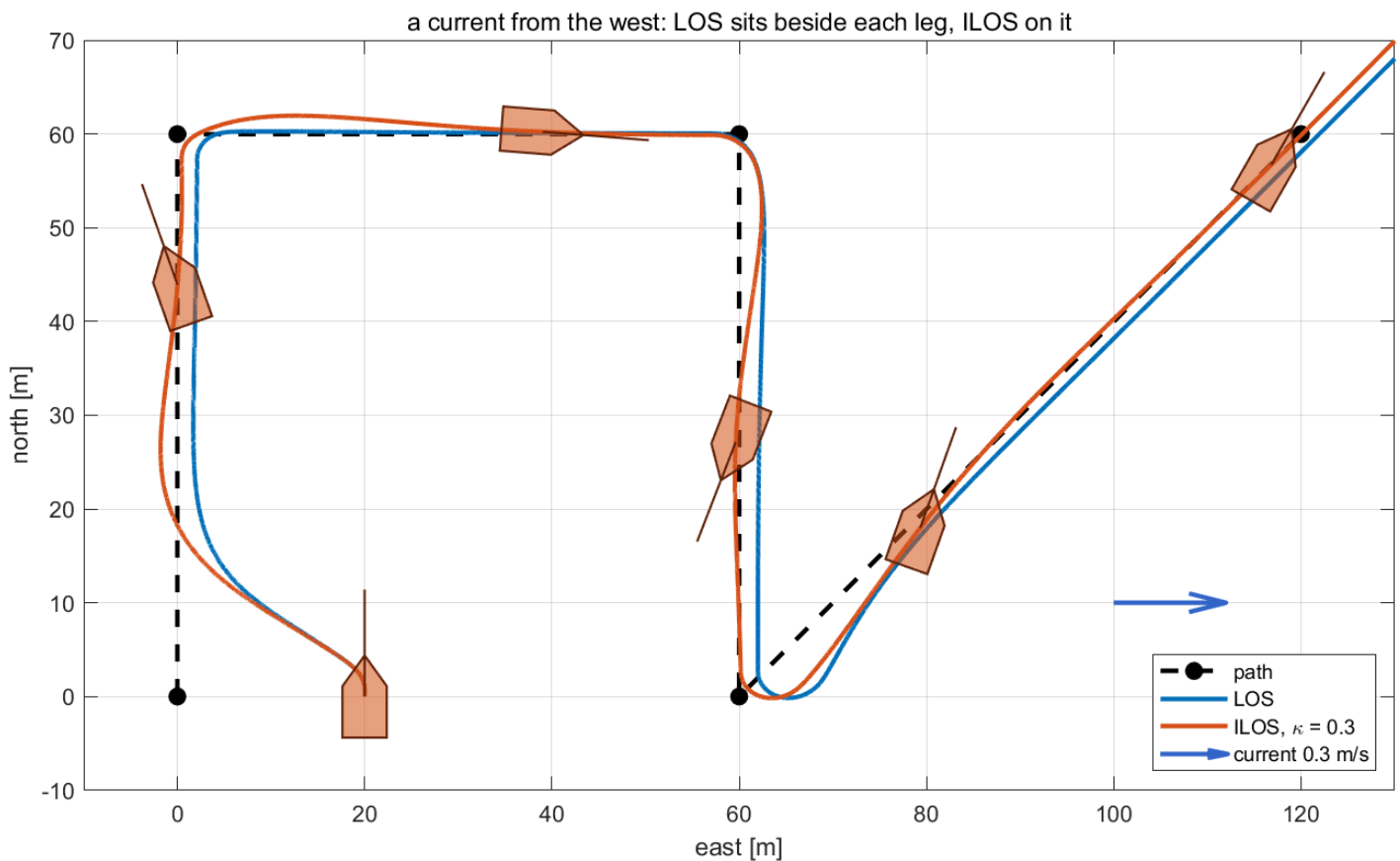
Opening and running.

```
W05_0_setup
V_c = 0.3; T_final = 450;           % 동쪽 조류 / a current to the east
open_system('W05_G_tuning')         % Run - kappa = 0.3 / the chosen value
kappa = 0;                           % Run - LOS 로 돌아간다 / back to LOS: it runs beside the path
W05_G_tuning_by_hand                 % 다섯 값을 다리별로 / all five values, leg by leg
```

Expected output:

W05 Experiment 5-6 the tuning order (Delta = 5 m, R_switch = 3 m, current 0.3 m/s east)

kappa	mean $ y_e $ on legs 1, 2, 3, 4 [m]				sum of legs 2-4 [m]	last waypoint at [s]
0	3.29	0.12	2.14	1.38	3.64	330.4
0.1	2.04	1.22	1.33	0.55	3.10	337.0
0.3	2.29	0.40	0.38	0.14	0.92	338.8
0.5	2.44	0.30	0.48	0.15	0.93	342.6
1	2.62	0.24	0.66	0.18	1.07	349.8



In the figure	Meaning
blue	LOS: beside each leg that crosses the current
orange with hulls	ILOS, $\kappa = 0.3$: on the legs; the hulls point into the current while the track stays on the path
arrow	the current, 0.3 m/s towards the east

Reading the figure and the table against the three steps.

Where to look	What is there	Which step or section it settles
the blue track beside legs 3 and 4	2.14 m and 1.38 m of mean error	§5-5 lines 2 to 4: the offset LOS must leave, now on a mission
the blue track on leg 2	0.12 m only	that leg runs with the current, so there is almost no crab angle to hold
the orange track on every leg	0.40, 0.38, 0.14 m	§5-5 line 7: the integral holds the lean instead
the hulls on the orange track	pointing into the current while the track runs along the leg	Week 4 §4-4: heading is not course, seen directly
the $\kappa = 0.1$ row	3.10 m, barely better than LOS	short legs: the value that was cleanest on 400 m has no time to act on 60 m
the last column, 330 → 350 s	the mission grows slower as κ grows	the integral's swing costs distance as well as accuracy
leg 1 in every row	2.0 to 3.3 m, and left out of the sum	it contains the 20 m start offset, which is a transient and not a following error

What the figure says

- The tuned ILOS follows the legs; the hulls show the crab angle it holds to do so.

What to try	What to watch
$\kappa = 0.3$; Run	the swing of Experiment 5-5 now happens at every corner: mean $ y_e $ after 60 s is 1.18 m against 0.64 m at $\kappa = 0.3$, and the last waypoint is reached at 385.9 s instead of 338.8 — both slower and less accurate
$V_c = 0$; $\kappa = 0.3$; Run	with no current there is nothing to find: 0.30 m against 0.20 m for LOS, and the same mission time. The integral is insurance, and it is not free

 In class

- Purpose** — close the week with the full order on a mission, and see Week 4 §4-4 in the picture.
- Point to** — the hull on the southbound leg, pointing west of south while the track runs due south.
- Ask** — "Why is κ chosen on the mission and not on the long leg of Experiment 5-5b?" Short legs leave little time to integrate; the gain that is best on 400 m is too slow on 60 m.
- Take away** — choose each parameter on the task it will face, in the order P, switching, I.

Part 2 · Laboratory run order

The experiments of Part 1 are worked through in order; this table is the index of what was run, for repeating the week at home.

Every example has **its own model**, laid out the same way: `guidance` → `heading autopilot` → `allocation` → `Otter`, each a commented MATLAB Function except the vessel. On the right, **one Scope** (top: cross-track error; bottom: commanded and actual heading) and an **XY Graph** that draws the track while the simulation runs.

Experiment	Model	Script	What it shows
5-0	all of them	<code>W05_0_setup</code> , <code>W05_1_build_guidance</code>	the parameters, and every model written from code
5-1	<code>W05_G_tuning</code>	—	where guidance sits, and the rotation that gives y_e
5-2	<code>W05_C_atan2</code>	<code>W05_C_aim_at_the_waypoint</code>	aiming at the next waypoint
5-3	<code>W05_D_LOS</code>	<code>W05_D_lookahead_distance</code>	LOS and the look-ahead distance
5-4	<code>W05_E_switching</code>	<code>W05_E_waypoint_switching</code>	the switching distance on a mission
5-5a	<code>W05_F_LOS</code>	<code>W05_F_LOS_in_a_current</code>	the offset a current leaves under LOS
5-5b	<code>W05_F_ILOS</code>	<code>W05_F_ILOS_removes_it</code>	the integral that removes it
5-6	<code>W05_G_tuning</code>	<code>W05_G_tuning_by_hand</code>	the tuning order on the mission in a current

- The scripts only repeat what Run already shows, for several values at once, and print the numbers quoted in Part 1.
- Each experiment ends with an **In class** note: purpose, what to point at, a question with its answer, and the sentence to take away.

Summary

Week Summary

Step	What was done	How it was verified
1	put a guidance law in front of the Week 4 autopilot	<code>check_overlaps</code> 0 in five models; y_e equals MSS <code>crosstrackWpt</code>
2	atan2 against LOS	Experiment 5-2: 10.75 m against 0.04 m halfway to the waypoint
3	Δ as the P gain $1/\Delta$	Experiment 5-3: 118 s at 40 m, swinging at 0.5 m; $\Delta = 5$ m
4	the switching distance	Experiment 5-4: worst corner 2.98 m at $R = 3$ m
5	a current and ILOS	Experiments 5-5a and 5-5b: LOS offset = $\Delta \tan(\text{heading})$ to the millimetre; ILOS removes it
6	the tuning order on the mission	Experiment 5-6: $\kappa = 0.3$, legs 2–4 summed 0.92 m against 3.64 m for LOS

Progress Check

★ Minimum condition for following Week 6

Theory

- ☐ Able to compute y_e from two waypoints and a position.
- ☐ Able to explain why LOS is a P controller and which way Δ moves its gain.
- ☐ Able to predict the LOS offset in a cross current from the heading held.
- ☐ Able to explain what the integral of ILOS does and why its denominator limits windup.

Laboratory

- ☐ `W05_1_build_guidance` ran and every model reported `overlapping lines: 0`.
- ☐ A parameter was changed in the Command Window and the XY Graph showed the new track.
- ☐ `W05_check(1)`, `(2)` and `(3)` pass on a model built from `W05_P1_start`.

In-class laboratory — build the guidance layer by hand

The second hour of the Week 5 session is spent building the guidance block in Simulink. A complete Week 4 vessel is provided with one input left open: the commanded heading.

```
cd lectures/W05_simulink/problems
W05_P1_start           % creates W05_P1.slx – a Week 4 vessel, no guidance
W05_check(1)          % run this whenever, as often as needed
```

	Problem	Time	The number it must reproduce
1	The LOS law on leg 1	25 min	$\pi_p = 0$ exactly; y_e from 18 m to 0; $\psi_d \rightarrow \pi_p$
2	atan2 against LOS, same vessel	20 min	LOS holds the line; atan2 does not
3	A current, and the offset that stays	15 min	settled $y_e = \Delta \tan$ of the crab angle

- The problem sheet is [W05_simulink/problems/README.md](#) , and reference answers are in [W05_simulink/solutions/](#) .
- Problem 3 ends where §5-5 continues: the integral that removes the offset.

Assignment 5

- **Due:** before the Week 6 session
- **Submit:** the modified [W05_0_setup.m](#) , the numbers requested below, and two figures

① Requirements

Double the legs of the mission (every waypoint coordinate times two, legs of 120 m) and tune Δ , R and κ again by the order of §5-6 in a 0.3 m/s current.

② Verification — mandatory

★ A claim is not a result. Numbers are required, with the averaging window stated

1. The table of Experiment 5-3 on one 240 m leg with the chosen Δ and its neighbours.
2. The table of Experiment 5-4 on the new mission.
3. The table of Experiment 5-6 on the new mission, and the κ chosen.
4. The LOS offset in the current, next to $\Delta \tan$ of the heading held.

③ Analysis (5–10 lines)

Compare the chosen κ with 0.3, and explain from the leg length why it moved in that direction.

Grading

Criterion	Weight
The model runs and produces the requested output	25%
Verification performed and numbers reported	40%
Correctness of the analysis	25%
Readability of the code	10%

Troubleshooting

Symptom	Cause	Fix
<code>W05_0_setup</code> stops in <code>mss_path</code>	the MSS toolbox is not at <code>Tools\MSS</code>	place MSS there; see the first page
the XY Graph shows nothing	the track is outside its axes	the axes are –20 to 140 m; a custom path needs <code>set_param(... '/track', 'xmax', ...)</code>
the vessel circles a waypoint	R smaller than the vessel can turn inside	raise <code>R_switch</code>
<code>guidance</code> complains that <code>WP_N</code> is undefined	<code>W05_0_setup</code> was not run	run it; the guidance block reads the waypoints from the workspace
a Scope does not match these notes	a changed variable is still in the workspace	run <code>W05_0_setup</code> again

References

Primary

- Fossen, T. I. *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed., Wiley, 2021, §12.3 (line-of-sight guidance and its integral form).
- Børhaug, E., Pavlov, A. and Pettersen, K. Y. "Integral LOS control for path following of underactuated marine surface vessels in the presence of constant ocean currents." *47th IEEE Conference on Decision and Control*, 2008, pp. 4984–4991. DOI: 10.1109/CDC.2008.4739352.
- MSS toolbox, `Tools\MSS\GNC/crosstrackWpt.m` and `ILOSpsi.m`.

In this course

- Week 2 §2-6 to §2-8 and §2-12: P, I and the tuning order. Week 4 §4-4: heading is not course.
- The previous, model-based version of this week — full derivations, Lyapunov stability and ALOS — is kept in `W05_simulink/_previous_version/`.

Next Week

Week 6 — Control Allocation. This week gave the vessel the square allocation of Week 4 and never questioned it. Week 6 asks what happens when there are more actuators than degrees of freedom, when a thruster saturates, and when one fails.